

BIO102 Help Session 5

Phylogenetic Trees

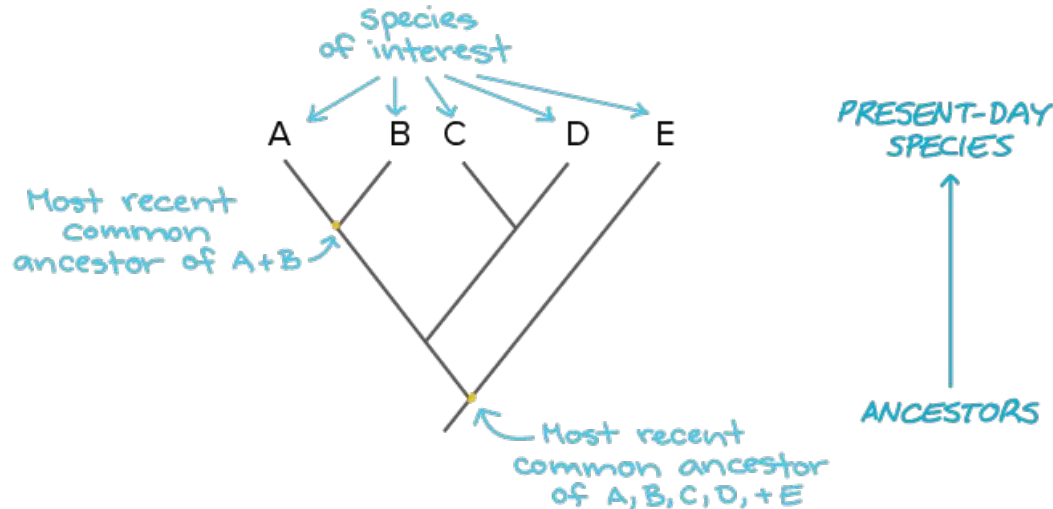
Phylogenetic Trees

- A **phylogenetic tree** is a diagram that represents evolutionary relationships among organisms. Phylogenetic trees are hypotheses, not definitive facts.
- The pattern of branching in a phylogenetic tree reflects how species or other groups evolved from a series of common ancestors.
- In trees, two species are **more related** if they have a more recent common ancestor and **less related** if they have a less recent common ancestor.
- Phylogenetic trees can be drawn in various equivalent styles. Rotating a tree about its branch points doesn't change the information it carries.

Phylogenetic Trees

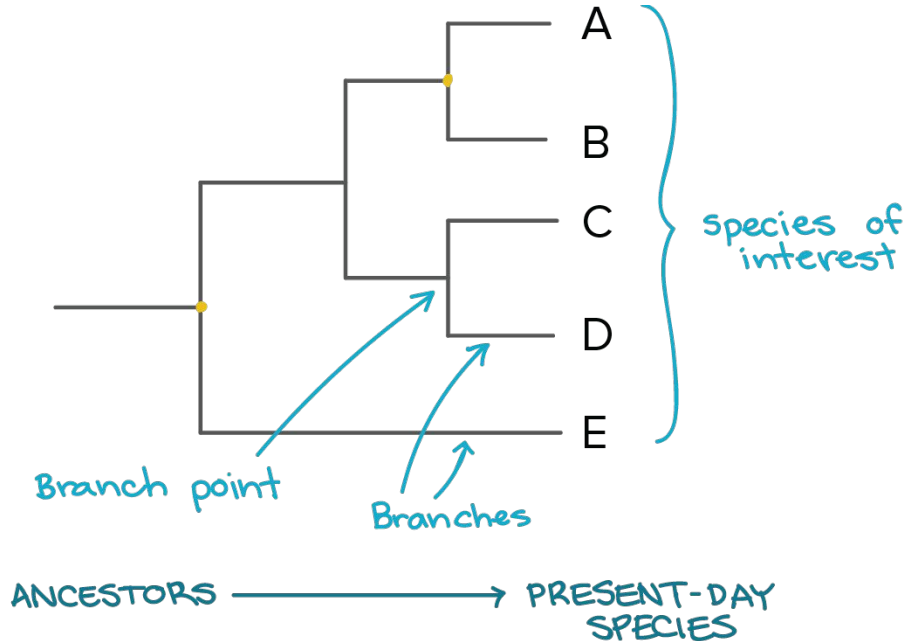
When we draw a phylogenetic tree, we are representing our best hypothesis about how a set of species (or other groups) evolved from a common ancestor.

This hypothesis is based on information we've collected about our set of species – things like their physical features and the DNA sequences of their genes.



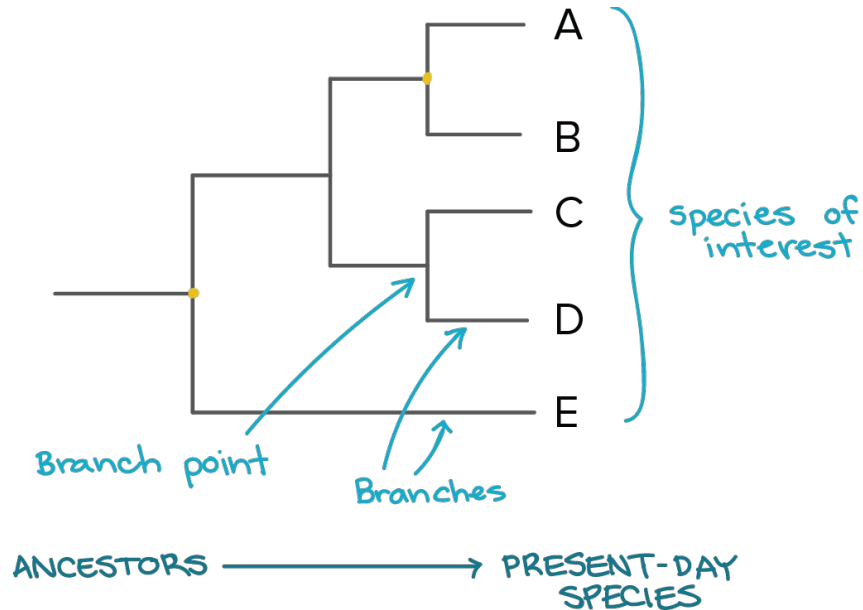
Anatomy of a Phylogenetic Tree

In a phylogenetic tree, the species or groups of interest are found at the tips of lines referred to as the tree's **branches**. For example, the phylogenetic tree below represents relationships between five species, A, B, C, D, and E, which are positioned at the ends of the branches:



Anatomy of a Phylogenetic Tree

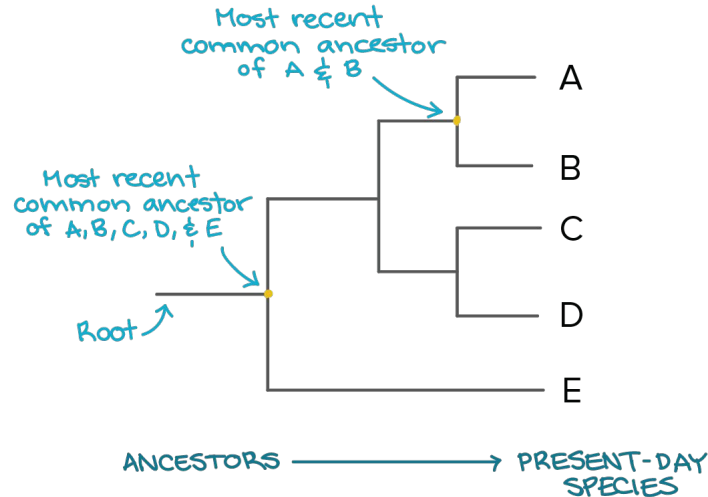
The pattern in which the branches connect represents our understanding of how the species in the tree evolved from a series of common ancestors. Each branch point (also called an **internal node**) represents a **divergence** event, or splitting apart of a single group into two descendant groups.



Anatomy of a Phylogenetic Tree

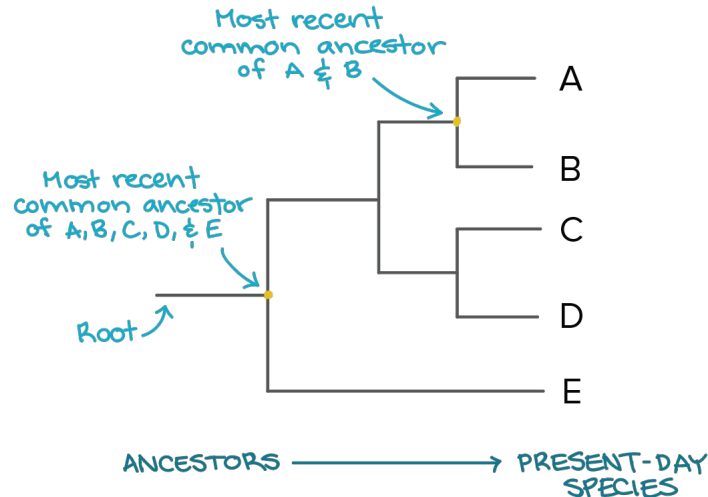
At each branch point lies the **most recent common ancestor** (MRCA) of all the groups descended from that branch point.

At the branch point right above the **root** of the tree, we would find the most recent common ancestor of all the species in the tree (A, B, C, D, E)



Anatomy of a Phylogenetic Tree

Each horizontal line in our tree represents a series of ancestors, leading up to the species at its end. For instance, the line leading up to species E represents the species' ancestors since it diverged from the other species in the tree. Similarly, the root represents a series of ancestors leading up to the most recent common ancestor of all the species in the tree.



Leaves / External
node/OTU

Edge/Branch/Lineage
In some trees,
Branch Length (Distance) ~
Time

Internal Node
(Common
Ancestor)

A and B are sister groups

taxon A

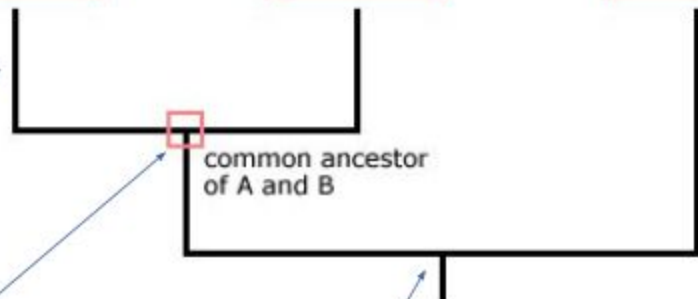
taxon B

C is the outgroup
to A and B

taxon C

common ancestor
of A and B

Root
(Common ancestor of all Operational
Taxonomic Unit- OTU)



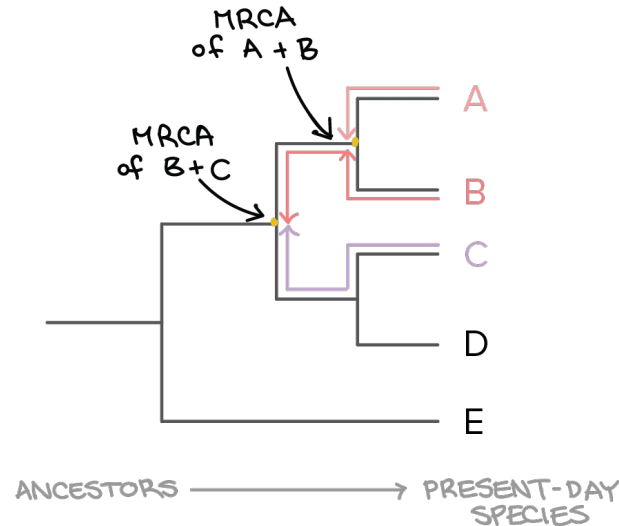
Which Species are More Related?

In a phylogenetic tree, the **relatedness** of two species has a very specific meaning. Two species are *more* related if they have a *more recent* common ancestor, and *less* related if they have a *less recent* common ancestor.

To find the most recent common ancestor of any pair or group of species, we start at the branch ends carrying the two species of interest and “walk backwards” in the tree until we find the point where the species’ lines converge.

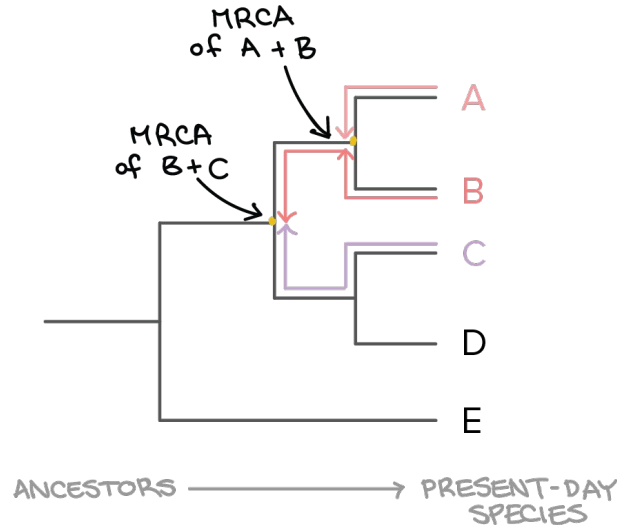
Which Species are More Related?

For instance, suppose that we wanted to say whether A and B or B and C are more closely related. To do so, we would follow the lines of both pairs of species backward in the tree. Since A and B converge at a common ancestor first as we move backwards, and B only converges with C after its junction point with A, we can say that A and B are more related than B and C.



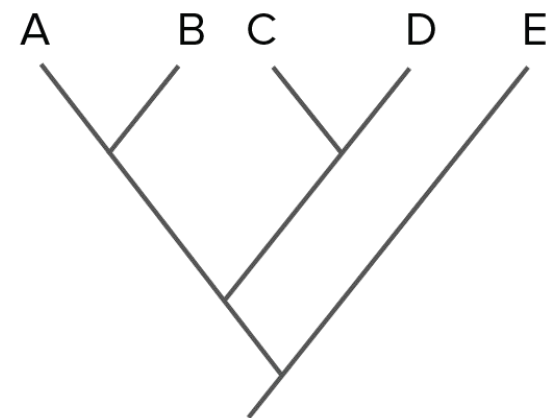
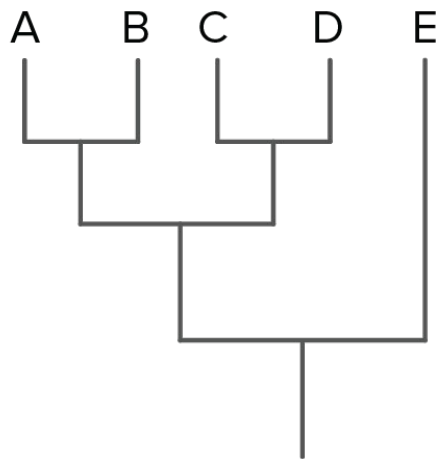
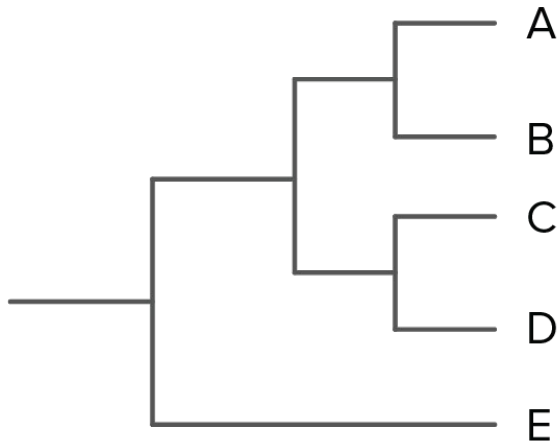
Which Species are More Related?

Importantly, there are some species whose relatedness we can't compare using this method. For instance, we can't say whether A and B are more closely related than C and D. That's because, by default, the horizontal axis of the tree doesn't represent time in a direct way. So, we can only compare the timing of branching events that occur on the same lineage (same direct line from the root of the tree), and not those that occur on different lineages.



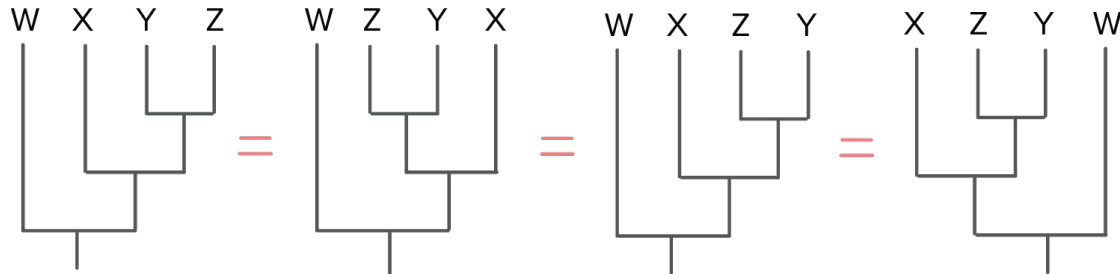
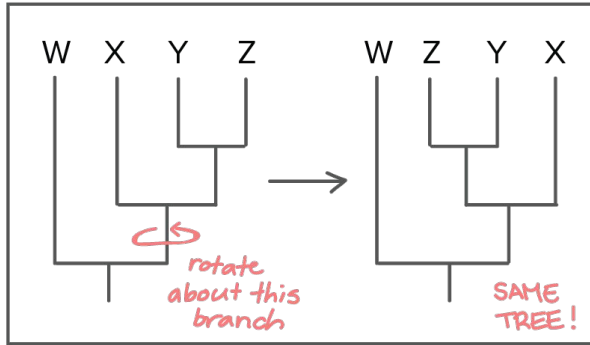
Reading Phylogenetic Trees

Phylogenetic trees are drawn in many different formats. The three trees above represent identical relationships among species A, B, C, D, and E. No branching patterns or recent-ness of common ancestors are different between the two trees. The identical information in these different-looking trees show that it's the branching pattern (and not the lengths of branches or the ordering of species) that's meaningful in a typical tree.



Reading Phylogenetic Trees

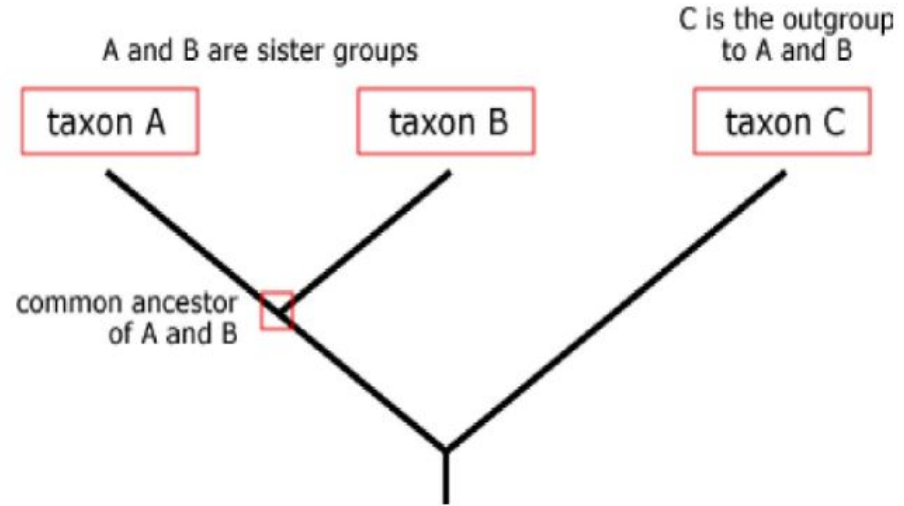
If you rotate the structures, using one of the branch points as a pivot, you don't change the relationships. All of the trees below show the same relationships among four species:



Outgroup

Many phylogenies also include an outgroup — a taxon outside the group of interest.

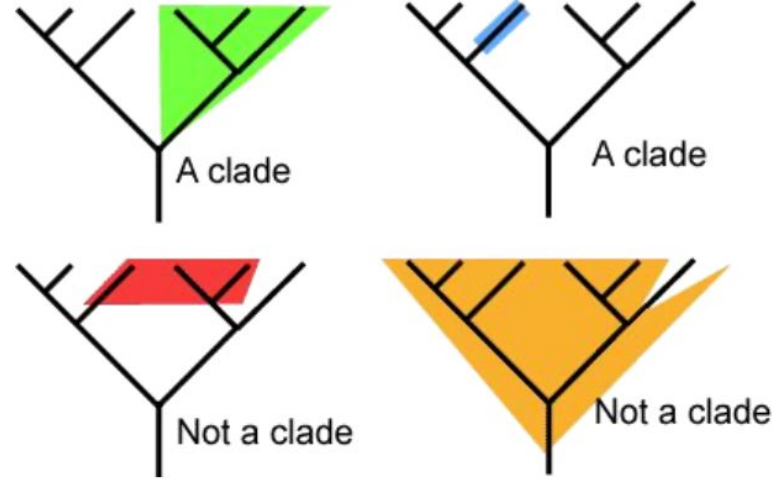
- All the members of the group of interest are more closely related to each other than they are to the outgroup. Hence, the outgroup stems from the base of the tree.
- An outgroup can give you a sense of where on the bigger tree of life the main group of organisms falls. It is also useful when constructing evolutionary trees.



CLADES

A clade is a group of organisms that are all descendent from a common ancestor; thus a clade includes an ancestor and all descendents of that Ancestor.

Clade can be thought of as a branch on the “tree of life”

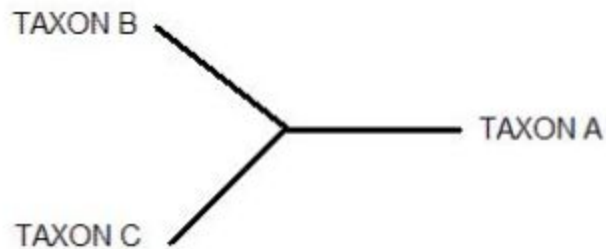
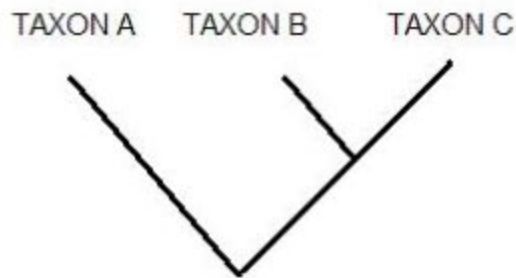


NESTED CLADES

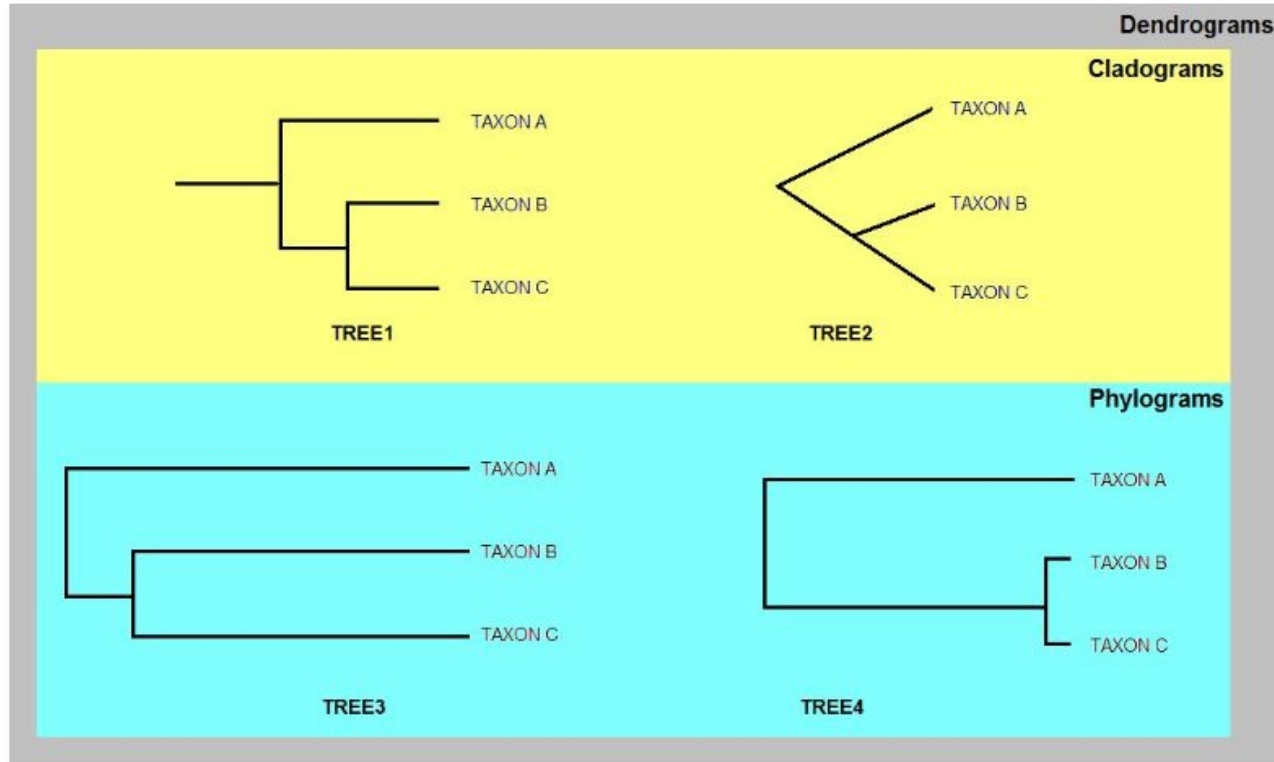


Clades are nested within one another — they form a nested hierarchy.

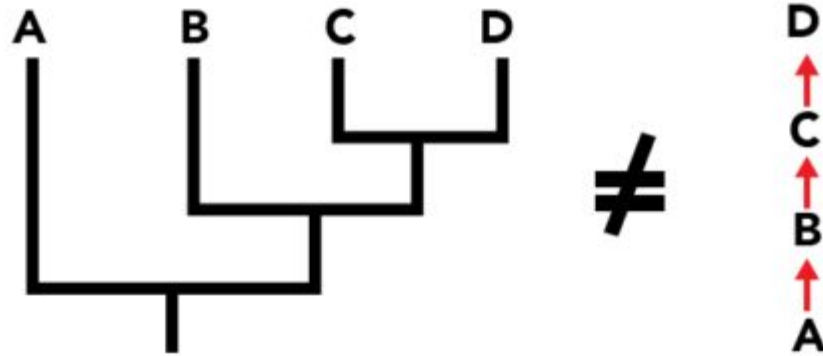
Types of trees: unrooted vs rooted



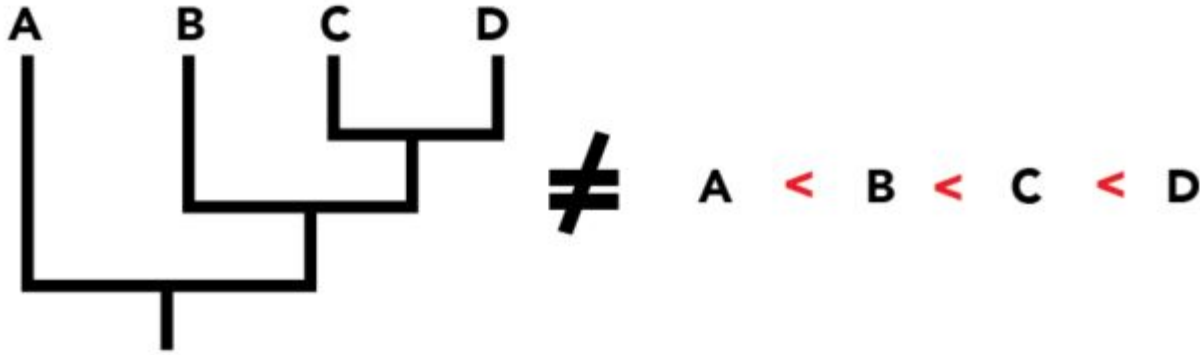
Dendrogram, cladogram, phylogram



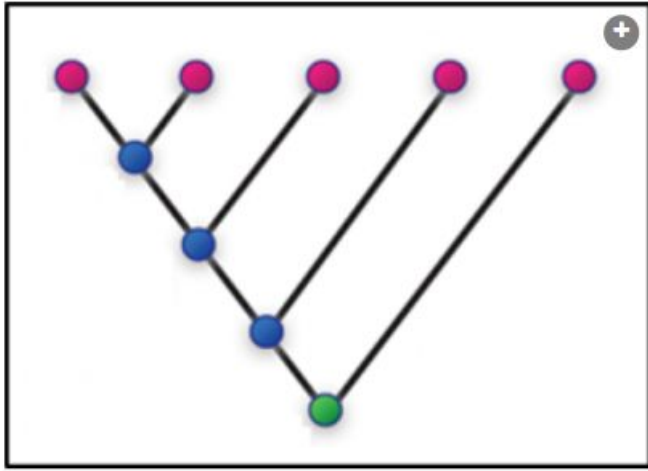
Misconceptions



Misconceptions

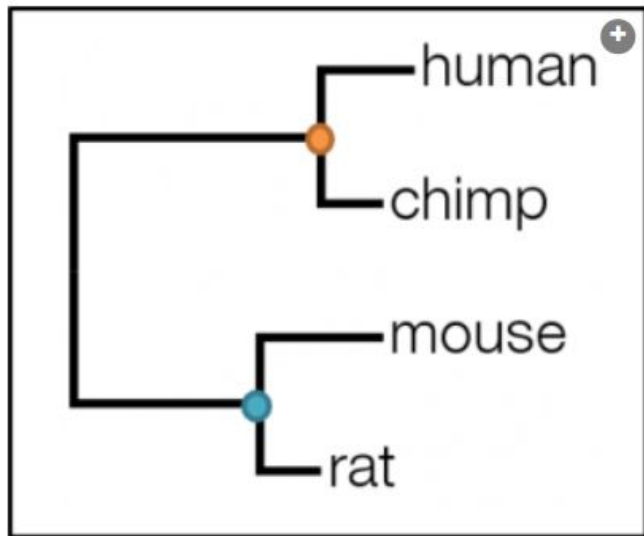


Questions



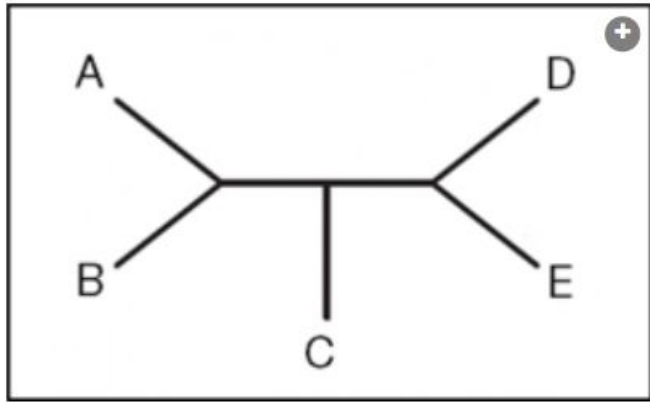
What are the types of nodes indicated with coloured spots in the tree above?

- ☐ Pink = root Blue = internal nodes Green = tips
- ☐ Pink = tips Blue = internal nodes Green = root
- ☐ Pink = internal nodes Blue = tips Green = root



The tree shown above is rooted. Which of the common ancestors occurred most recently?

- ☐ The ancestor of human, chimpanzee, mouse and rat
- ☐ The ancestor of human and chimpanzee
- ☐ The ancestor of mouse and rat



The tree above is unrooted. Where could you place the root so that A and B are not each other's closest relatives?

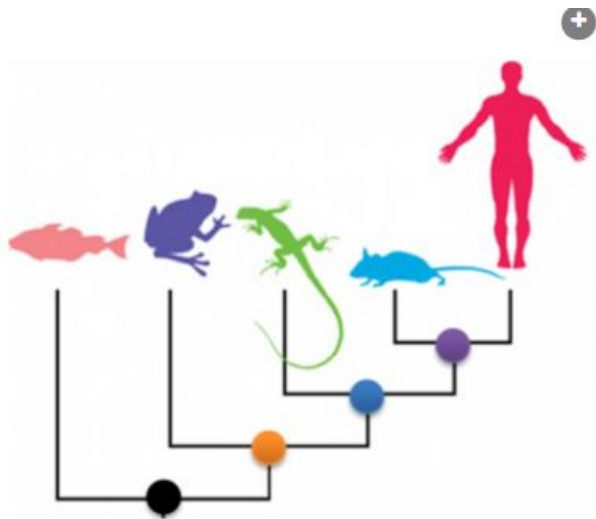
☐ At the node where D joins E

☐ At the node where A joins B

☐ At the midpoint of the tree

What is the typical unit that we use when describing estimates of evolutionary rate?

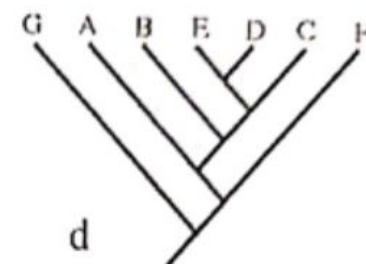
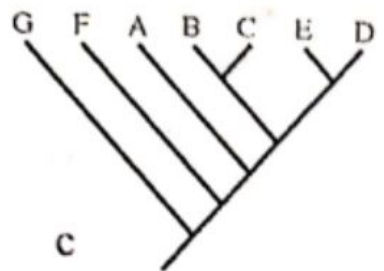
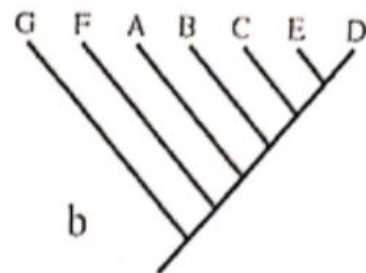
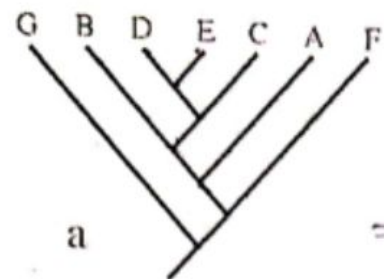
- ☐ mutations/site/year
- ☐ mutations/site
- ☐ substitutions/site
- ☐ substitutions/site/year



In the picture above, which species is most closely related to fish?

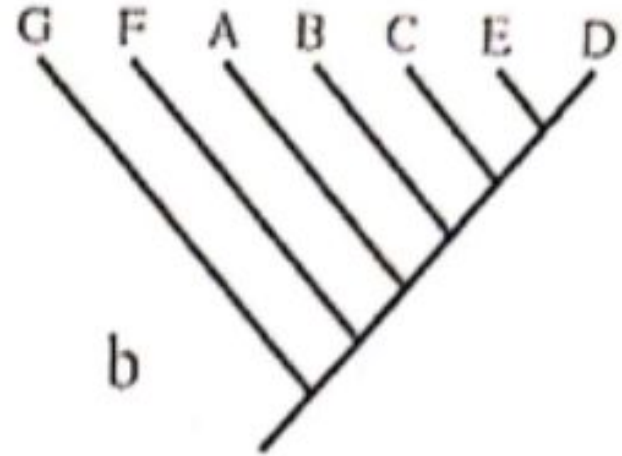
- ☐ lizard
- ☐ human and lizard are equally related to fish
- ☐ human

(ii) Consider the phylogenetic trees below. Which of the trees above shows a different relationship among the taxa than others? Explain very briefly. (2 marks)

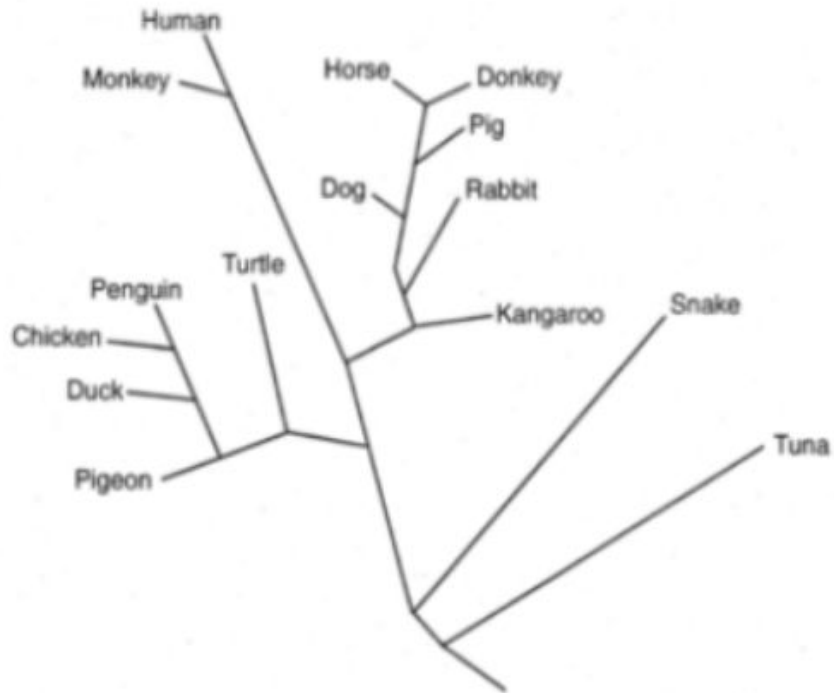


Comment on the statements below:

1. A is equally related to F and B
2. G is more similar to the common ancestor of all the organisms depicted in the phylogeny than E.



Consider the following phylogenetic tree and answer questions 8, 9, and 10.



8. According to this tree, dog is more closely related to which of these?
Mark)

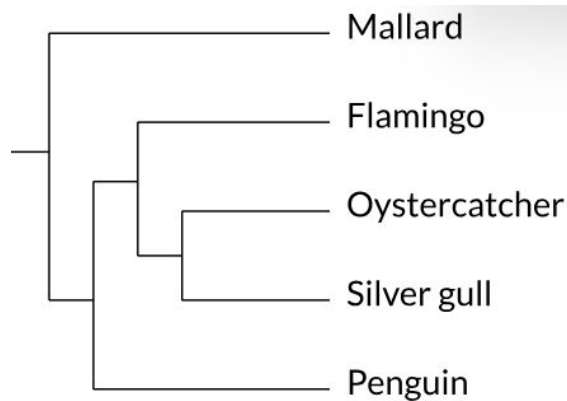
- ☐ Pig
- ☐ Rabbit
- ☐ Donkey
- ☐ Kangaroo

Consider the following phylogenetic tree and answer questions 8, 9, and 10.



10. According to this tree: (2 Marks)

- ☐ Snake and tuna are most similar to the common ancestor
- ☐ The lineage of mammals (kangaroo, rabbit etc including human and monkey) have undergone speciation the most number of times
- ☐ Human and tuna share a common ancestor
- ☐ Turtle, penguin, chicken, duck and pigeon form a clade

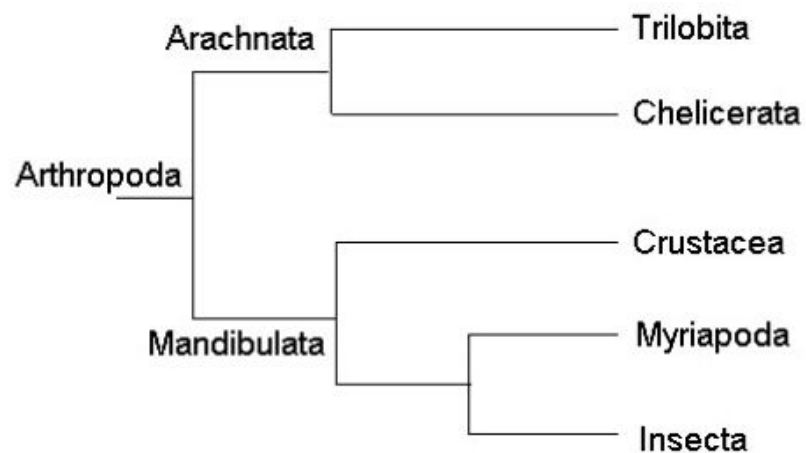


According to this tree, oystercatchers are more closely related to flamingos than to penguins.

Which of the following statements provides the best justification for this relationship?

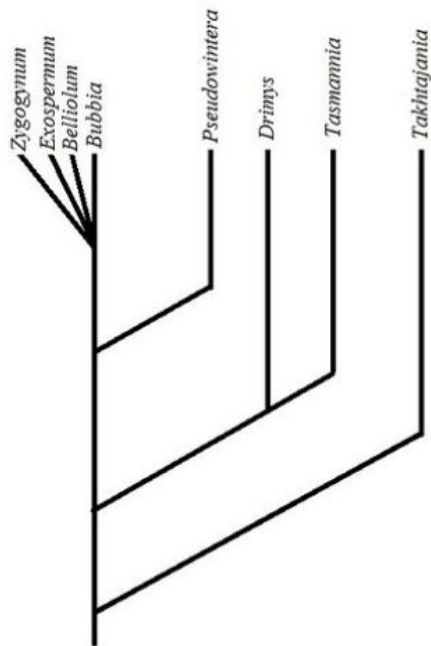
- ☐ (A) Oystercatchers share a more recent common ancestor with flamingos than with penguins.
- ☐ (B) Oystercatchers share a common ancestor with flamingos but not with penguins.
- ☐ (C) Both oystercatchers and flamingos are less closely related to the outgroup than penguins are.
- ☐ (D) The branch of the cladogram with oystercatchers is closer to the branch with flamingos than to the branch with penguins.

2. According to the phylogenetic tree, which organisms are the least related to each other evolutionarily?



- ☐ Trilobita and Insecta
- ☐ Insecta and Myriapoda
- ☐ Insecta and Crustacea
- ☐ Trilobita and Chelicerata
- ☐ Crustacea and Myriapoda

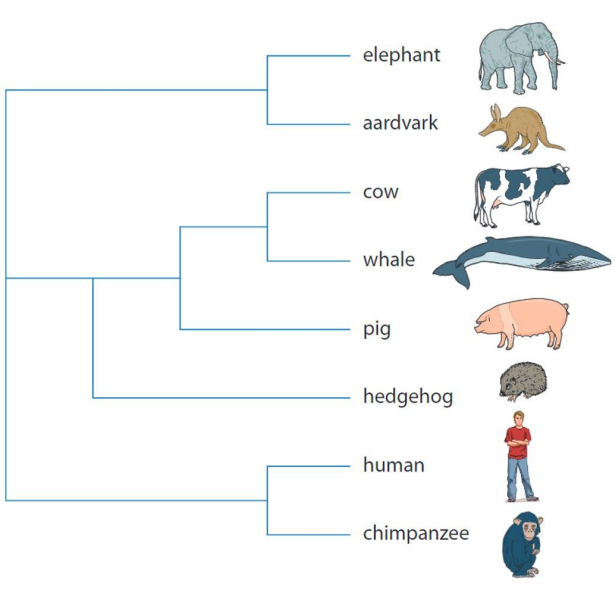
1. According to the cladogram shown, which organisms are most closely related?



- ☐ Exospermum and Takhtajania
- ☐ Exospermum and Tasmannia
- ☐ Belliolium and Bubbia
- ☐ Belliolium and Drimys
- ☐ Exospermum and Drimys

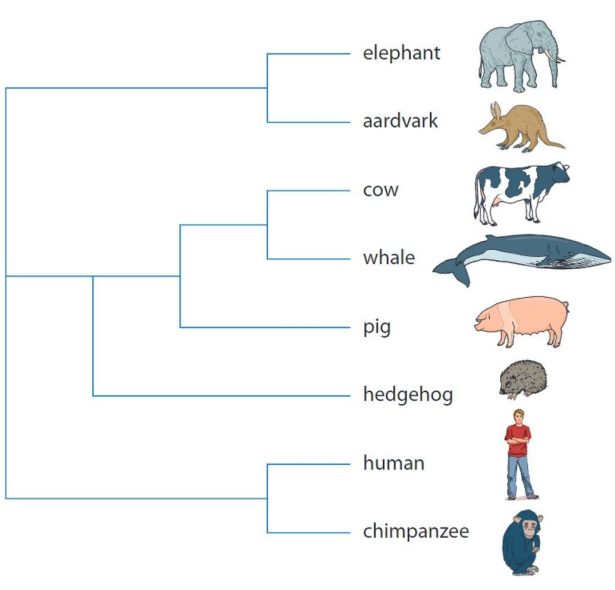
Based on the given tree diagram, which of the following statements is true?

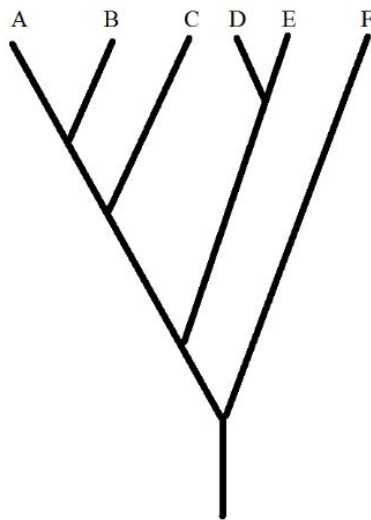
- A** Humans are closely related to elephants
- B** Cows are more closely related to pigs than whales
- C** Hedgehogs are closely related to armadillos
- D** Humans are closely related to chimpanzees



Based on the given tree diagram, which of the following statements is true?

- A** The cow is more closely related to the pig than to the whale
- B** Pig and hedgehog do not share any common ancestor
- C** Elephants are more closely related to aardvark than whales
- D** Pigs and chimpanzees are more closely related than humans





From the diagram above, to which species is species C most closely related?

D and E

All of the given species are equally related

D only

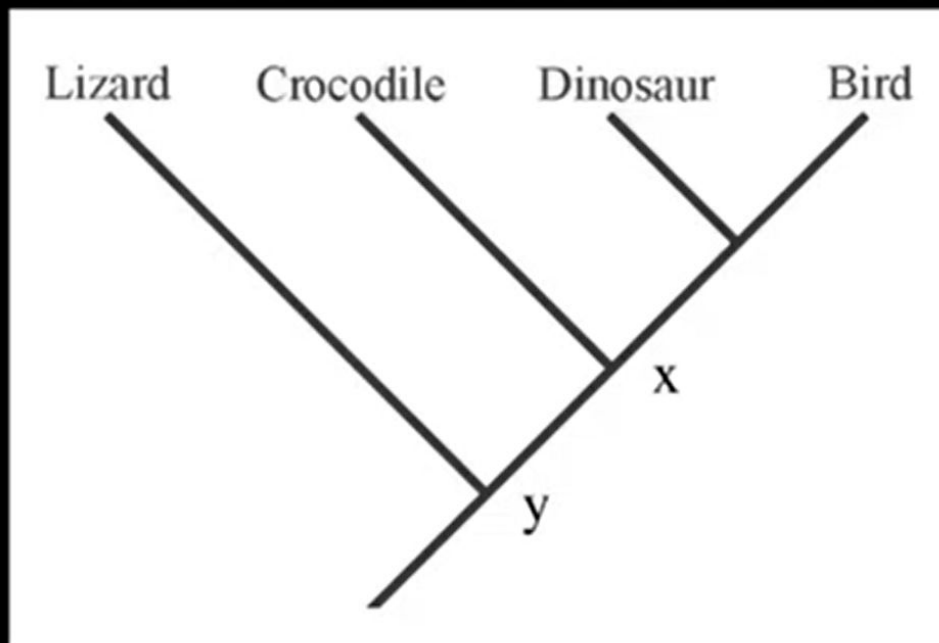
A and B

How does a phylogenetic tree relate to the passing of time?

- a) A phylogenetic tree relates to the passing of time because species branch off from each other at regular time intervals.
- b) A phylogenetic tree is not related to the passing of time because speciation is based on geographic changes.
- c) The phylogenetic tree only shows the order in which things took place.
- d) A phylogenetic tree relates to the passing of time when the diagram also shows how long ago the divergence from the common ancestor occurred.

By reference to the tree below, which of the following is an accurate statement of relationships?

- A) A crocodile is more closely related to a lizard than to a bird
- B) A crocodile is more closely related to a bird than to a lizard
- C) A crocodile is equally related to a lizard and a bird
- D) A crocodile is related to a lizard, but is not related to a bird



Assuming that in a phylogenetic tree of dogs, the red wolf has a longer branch length than the Coyote. Which of the following statements could be true?

- A** Evolution of the red wolf took more time
- B** The evolution of the Coyote took more time
- C** The evolution of the red wolf is incomplete
- D** The evolution of the red wolf is quicker than Coyote

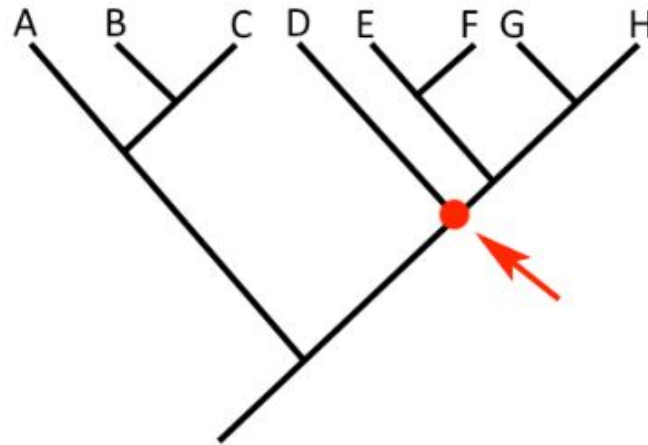
2. If cladistics were used to build a phylogenetic tree of cats, this would be the most appropriate outgroup

(a) tiger

(b) leopard

(c) wolf

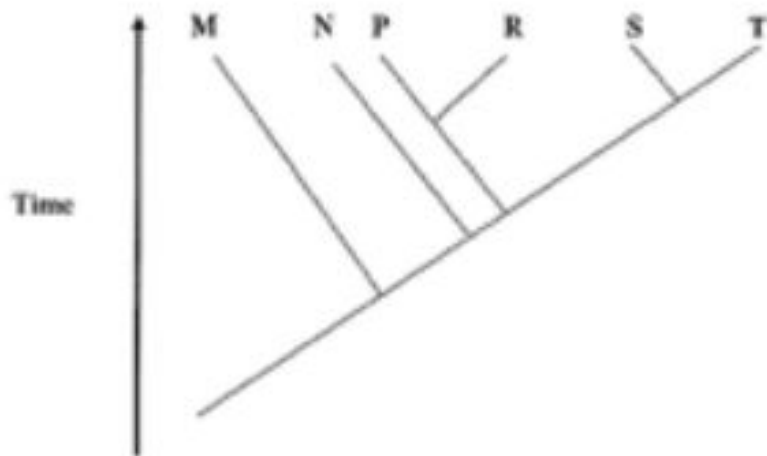
(d) domestic cat



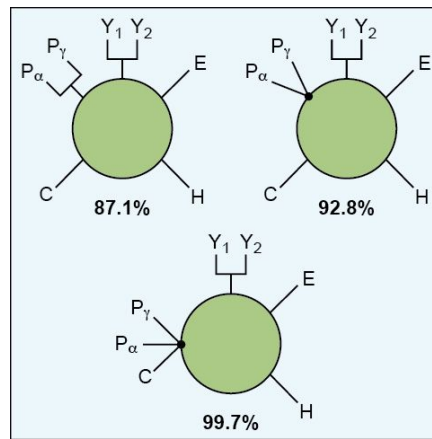
Question

In this phylogenetic tree, the identified node represents the shared common ancestor of which taxa?

Consider the following phylogeny:



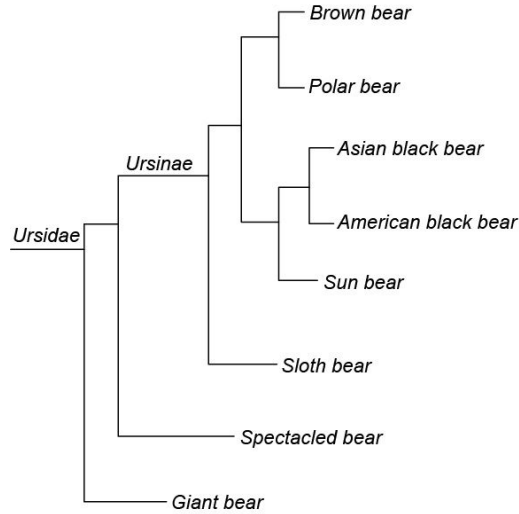
How many clades is species "S" a member of?



The images show various models that explain the phylogenetic relationships between two yeasts (Y1 and Y2), two protobacteria (Pg and Pa), a cyanobacterium (C) and three other bacteria (H, E, S). Scientists have constructed these models based on the genomes of these microorganisms.

Which statement about these models is correct?

- a.** It is most likely that Pa and Pg share a common ancestor. That common ancestor in turn shares a common ancestor with C.
- b.** Y1 and Y2 descended from a common ancestor in every model.
- c.** In the most likely model, E and H share a common ancestor. That common ancestor in turn shares a common ancestor with C.
- d.** In the least likely model, C and Pa are closely related.

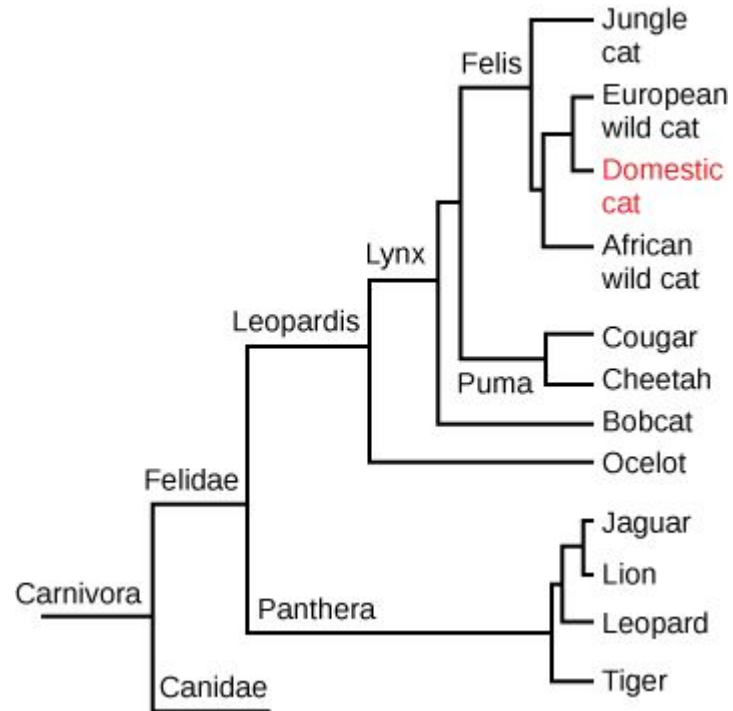


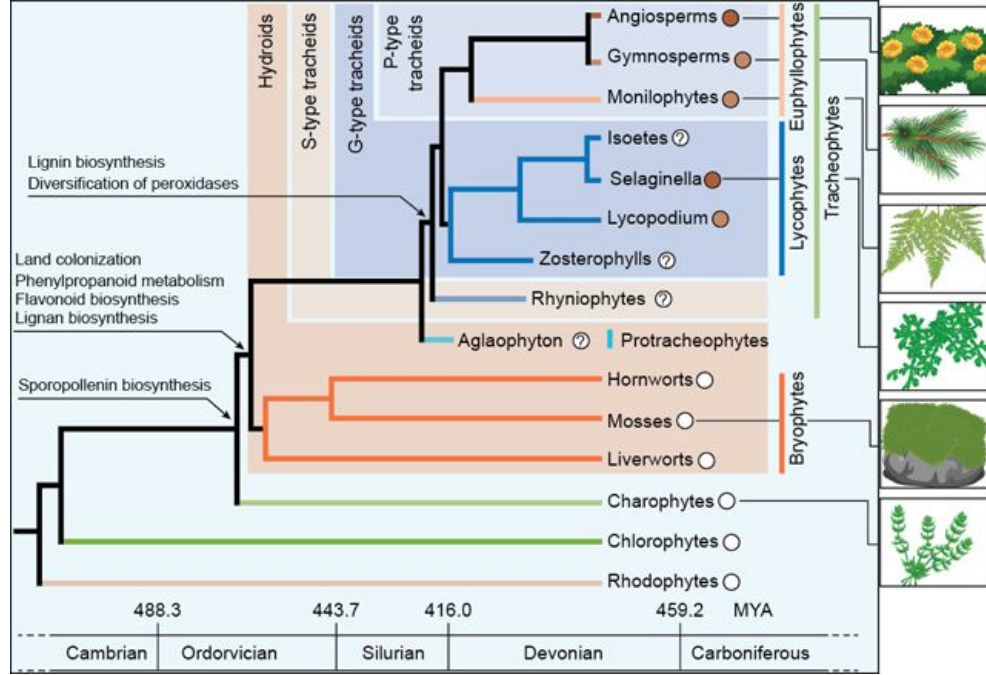
One of the phylogenetic trees proposed for bears (family Ursidae) is shown in the image.

What are the two oldest species in this family according to this tree?

- a.** The brown bear (grizzly) and the polar bear.
- b.** The Asian and American black bears.
- c.** The sun ber and the sloth bear.
- d.** The spectacled bear and the giant pands.

Based on the phylogenetic tree below, is the Jungle cat likely closer related to a tiger or a cougar? Why?

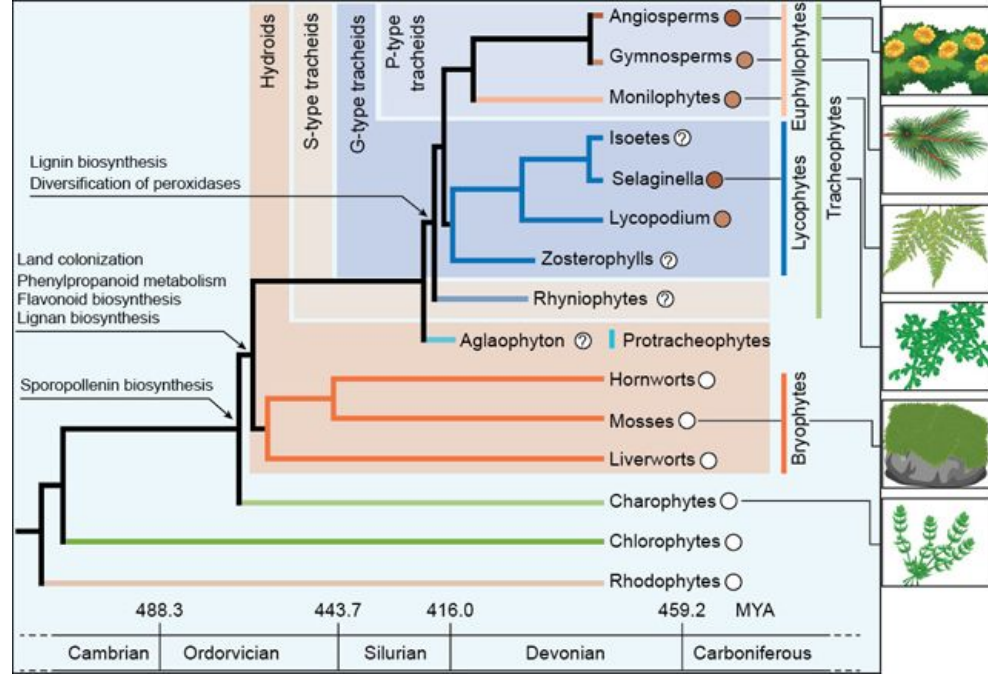




The phylogenetic tree shows the relationship between plants.

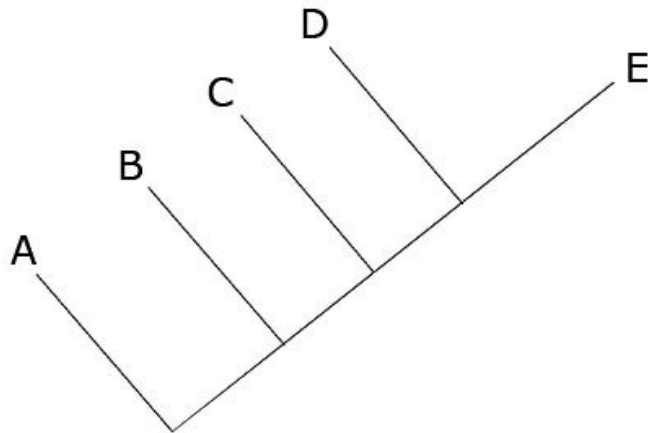
What is the closest relative of hornworts?

- a. Liverworts.
- b. Mosses.
- c. Protracheophytes.
- d. Charophytes.



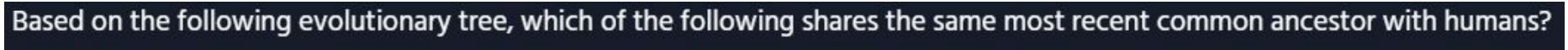
What is a trait shared with the common ancestor of charophytes (a group of algae) and liverworts (a group of non-vascular plants)?

- The common ancestor has colonized the land.
- The common ancestor could synthesize phenylpropanoids (a group of important organic molecules).
- The common ancestor could synthesize lignin (the major component of bark).
- The common ancestor could synthesize sporopollenin (the tough outer covering of plant spores).

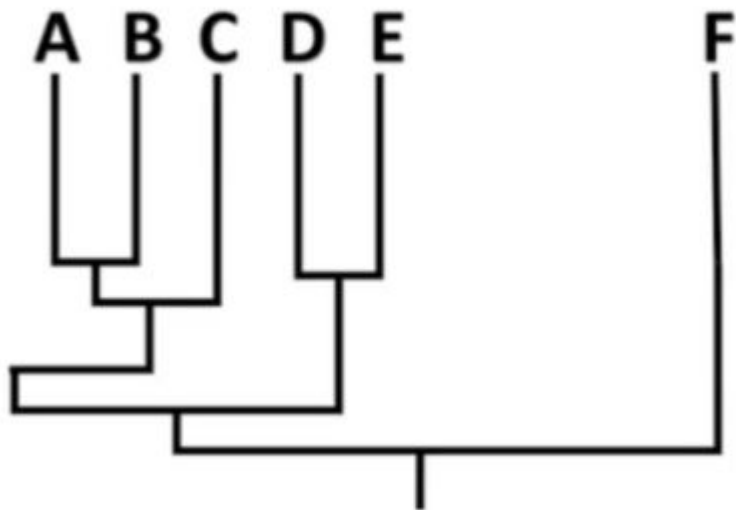


The tree above shows the phylogenetic relationships between four species. A scientist wishes to perform a genetic analysis on all four species in which she determines the number of genetic similarities between all four species. What would she most likely find regarding the genetic similarities between species A, B, D and E?

- a.** Species D and E would share more genetic similarities with each other than with species A and B, and vice versa.
- b.** Species A and E would share more genetic similarities with each other than with species B and D, and vice versa.
- c.** Species D and A would share more genetic similarities with each other than with species A and B, and vice versa.
- d.** Species D and B would share more genetic similarities with each other than with species A and E.



Which of the following is considered an outgroup?



A

Species A

B

Species D

C

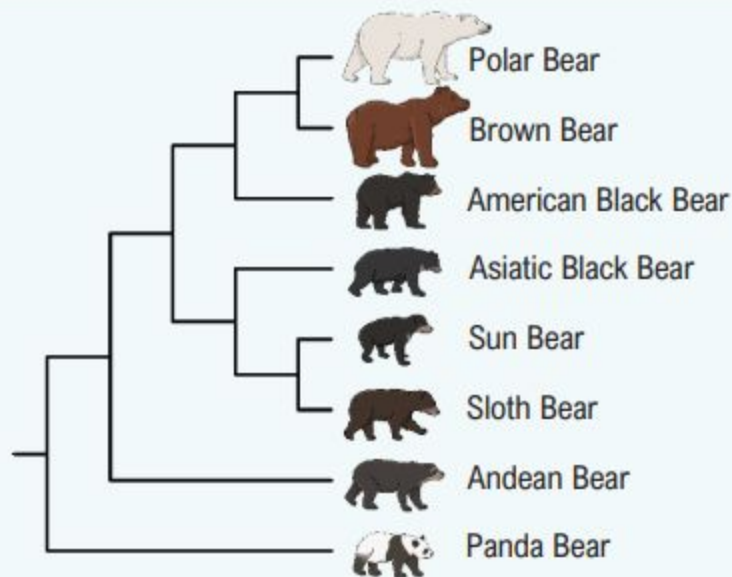
Species F

D

Species B

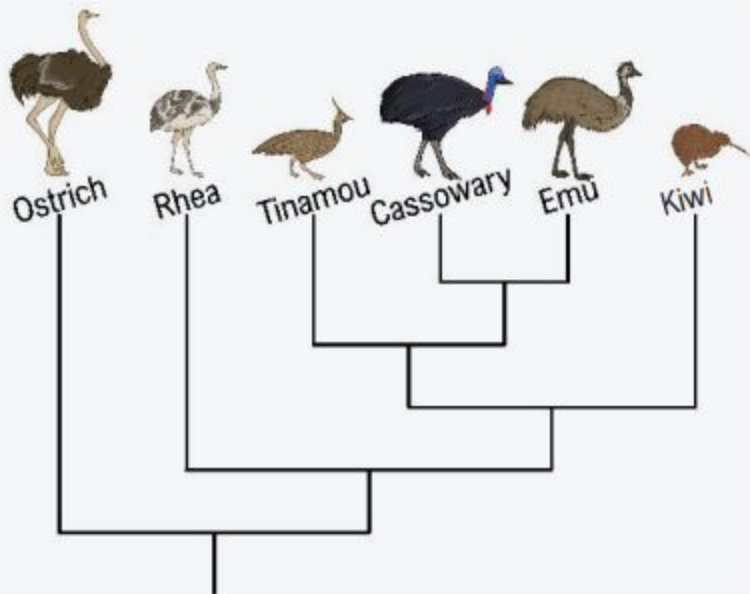
Evaluate the following statements as true or false based on the following phylogenetic tree.

- ____ This tree shows that polar bears have undergone the most evolution.
- ____ The sun bear is most closely related to the sloth bear.
- ____ The American black bear and Asiatic black bear are most closely related to each other.
- ____ The Andean bear and the panda bear are most closely related to each other.
- ____ The brown bear is equally related to the Asiatic black bear and the sloth bear.
- ____ This tree shows that the ancestor of all bears was most similar to a panda bear.
- ____ The American black bear is more closely related to a polar bear than it is to an Asiatic black bear.



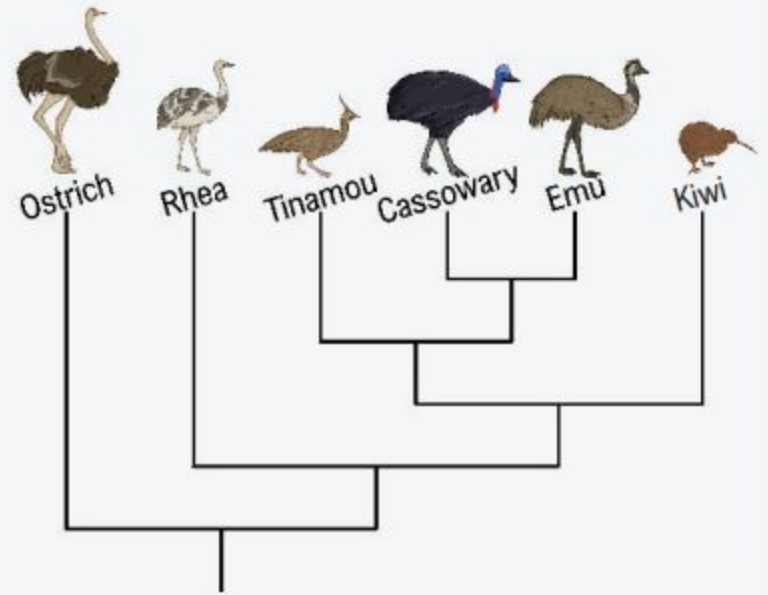
Based on the tree below, which of the organisms listed is most closely related to the tinamou?

- a) Rhea.
- b) Kiwi.
- c) Emu.
- d) Ostrich.



According to the phylogenetic tree below, which statement is correct?

- a) The rhea is most closely related to the tinamou.
- b) The rhea is equally related to both the cassowary and the kiwi.
- c) The ostrich and the rhea are most closely related to each other.
- d) The kiwi is more closely related to the rhea than it is to the cassowary.



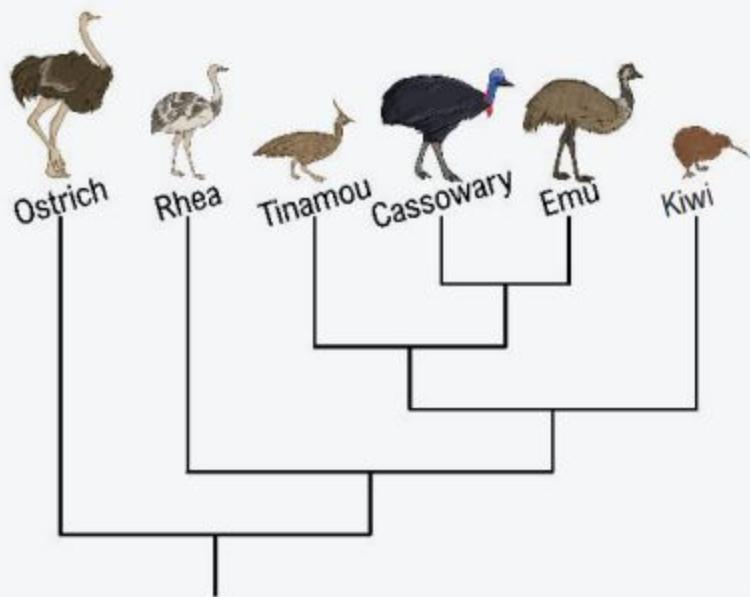
Based on the tree below, which of the birds listed is most distantly related to the tinamou?

a) Cassowary.

b) Kiwi.

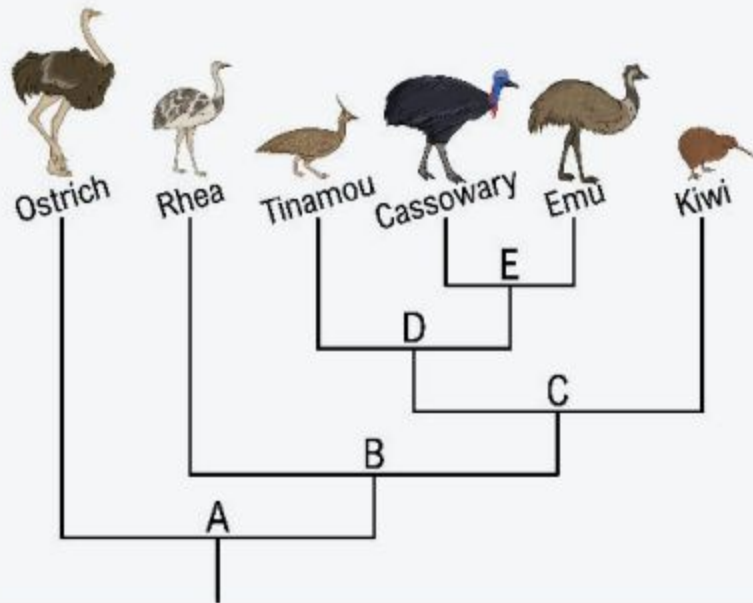
c) Rhea.

d) Emu.



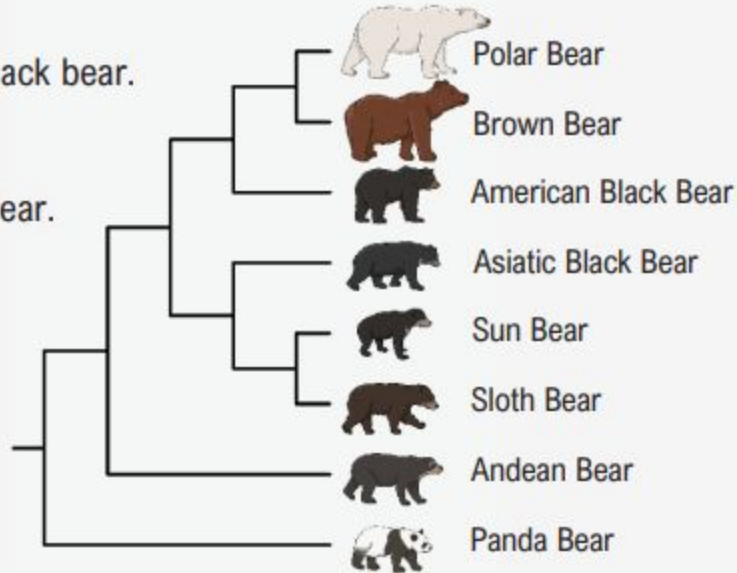
Which node represents the most recent common ancestor of the rhea and the kiwi?

- a) A.
- b) B.
- c) C.
- d) D.
- e) E.

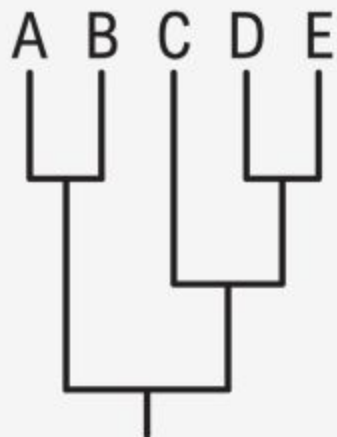


Which group represents a clade according to the tree below?

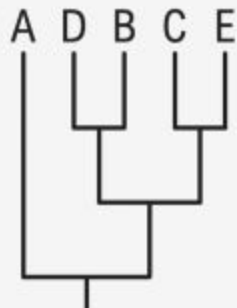
- a) Asiatic black bear & sun bear.
- b) Asiatic black bear & American black bear.
- c) Panda bear & Andean bear.
- d) All the bears except the panda bear.



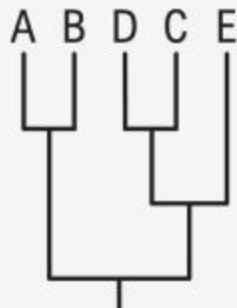
Based on the tree below, which tree shows the same evolutionary relationships?



a)



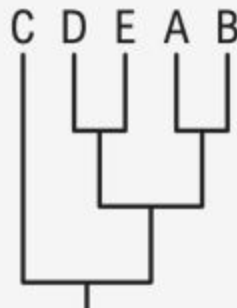
b)



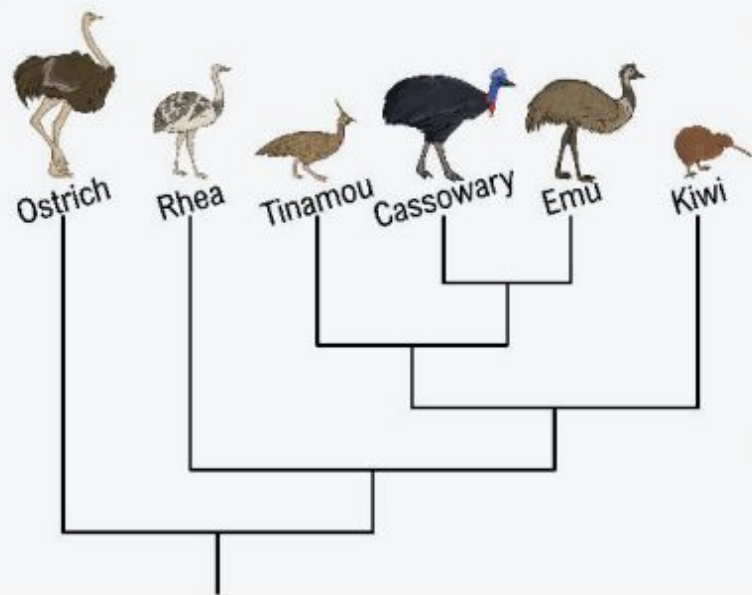
c)



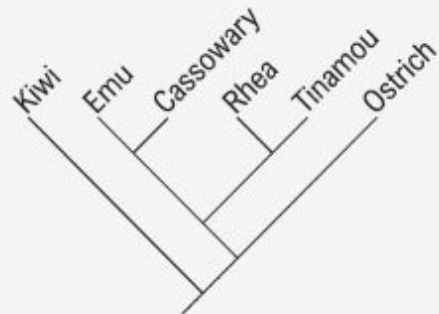
d)



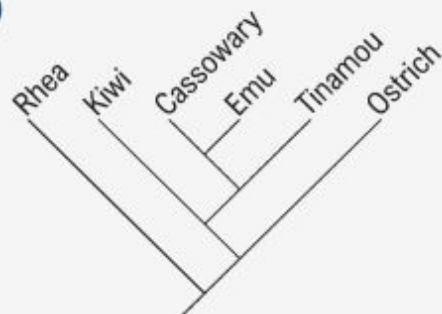
Based on the tree below, which tree shows the same evolutionary relationships?



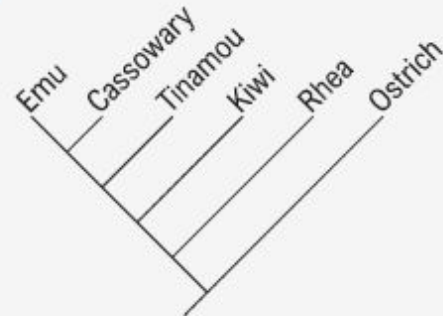
a)



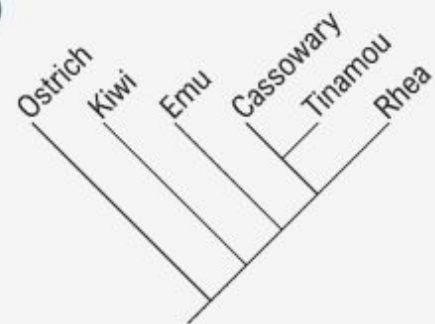
b)



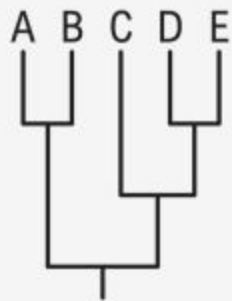
c)



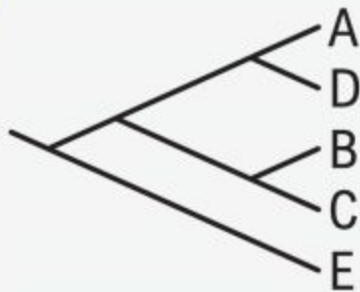
d)



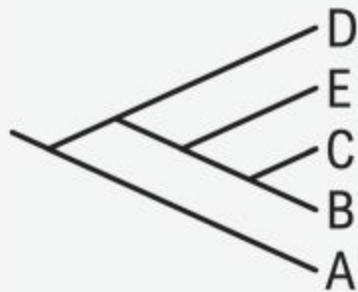
Which tree below shows the same evolutionary relationships?



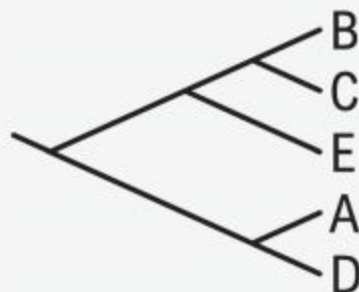
a)



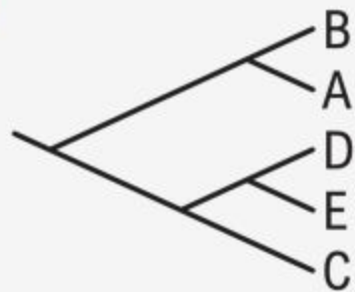
b)



c)



d)

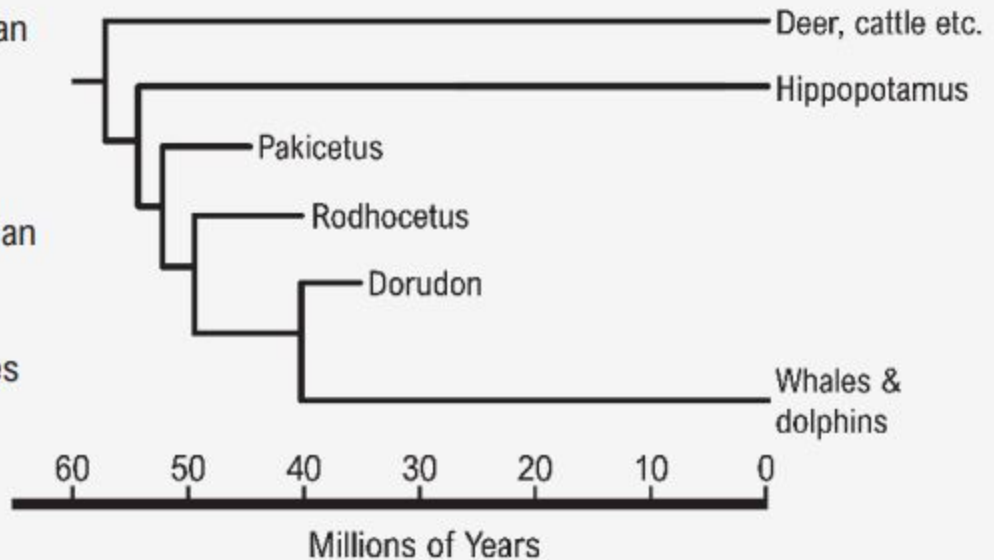


In a tree that shows the amount of evolution, what does a shorter branch length indicate?

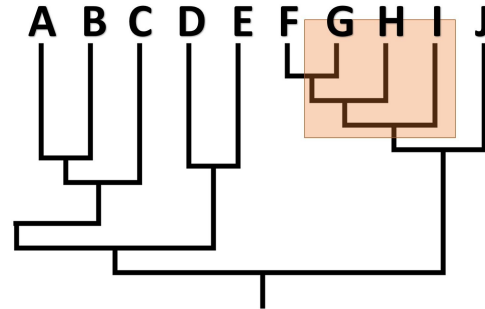
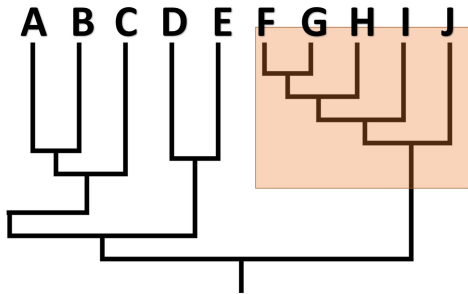
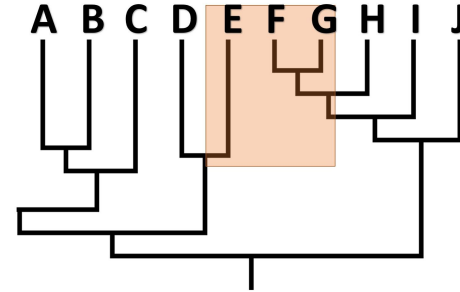
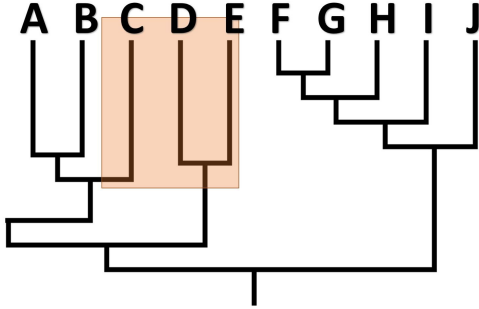
- a) Earlier branching events.
- b) Fewer genetic changes.
- c) Extinction events.
- d) Shorter time periods.

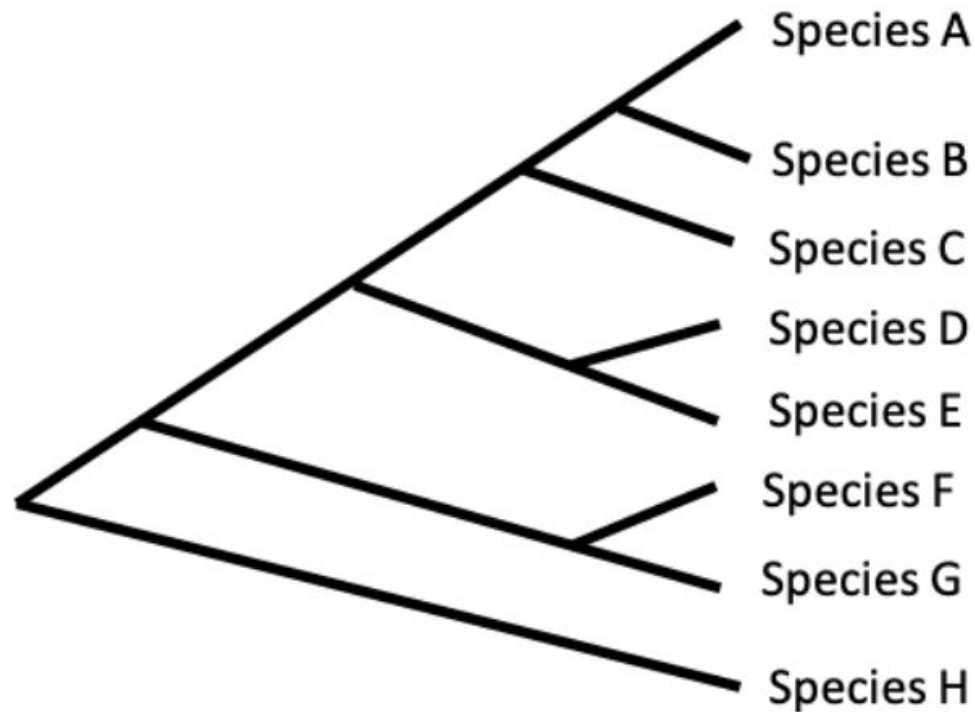
From the phylogenetic tree below, what can you infer from the branch lengths for the organisms Pakicetus, Rodhocetus, and Dorudon?

- a) They are more closely related to each other than to other organisms on the tree.
- b) They are extinct taxa.
- c) Those taxa experienced less genetic change than other branches on the tree.
- d) They represent the ancestors of modern whales and dolphins.



Which of the following images is considered a clade?

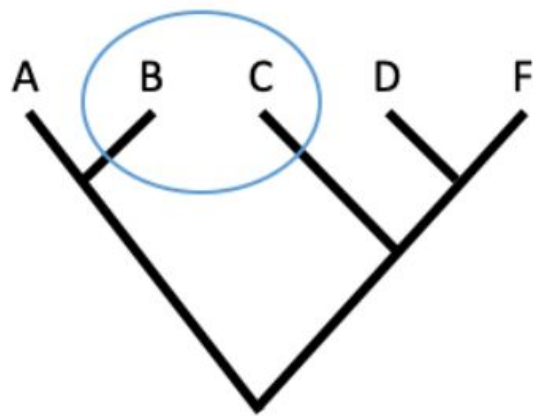




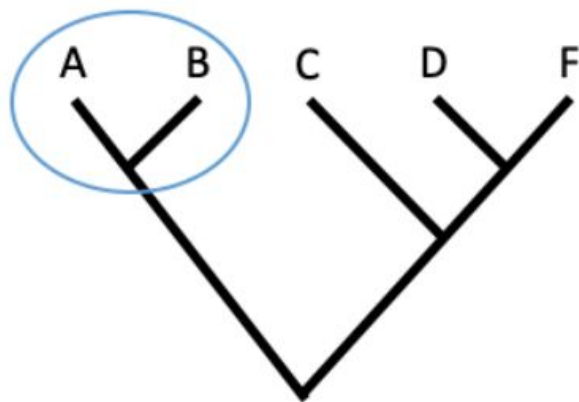
Label the follow statements True or False

- A. Species B is more closely related to Species D and E than Species F and G
- B. Species C is more closely related to Species D than E
- C. Species H is more closely related to Species G and Species A

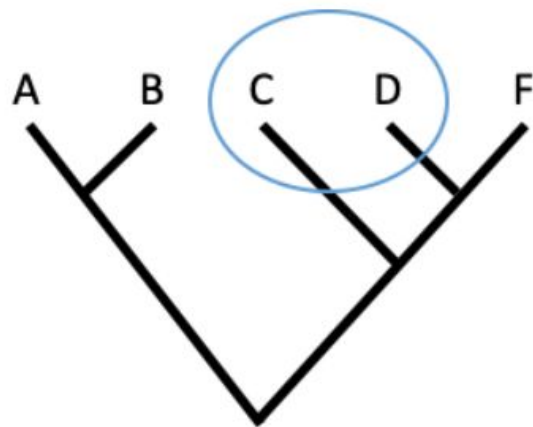
Which of the following trees has a monophyletic group circle on it?



Tree 1

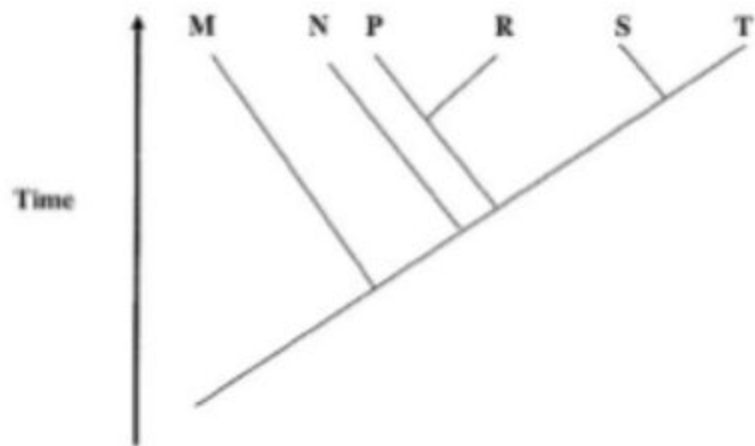


Tree 2

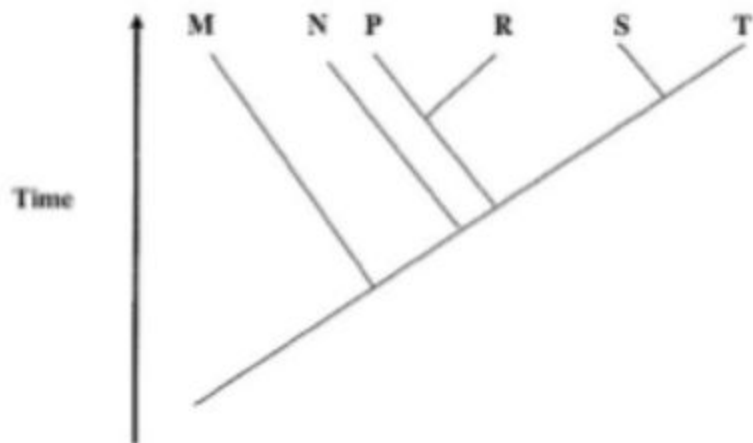


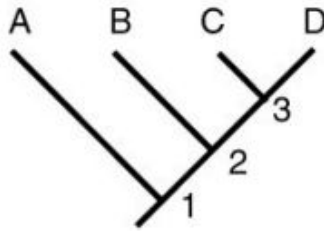
Tree 3

Concerning the phylogeny, which species is most distantly related to N?

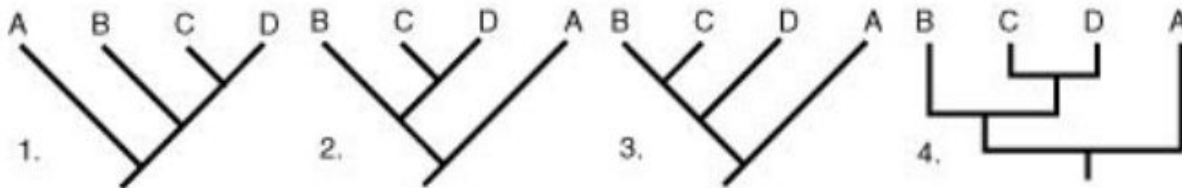


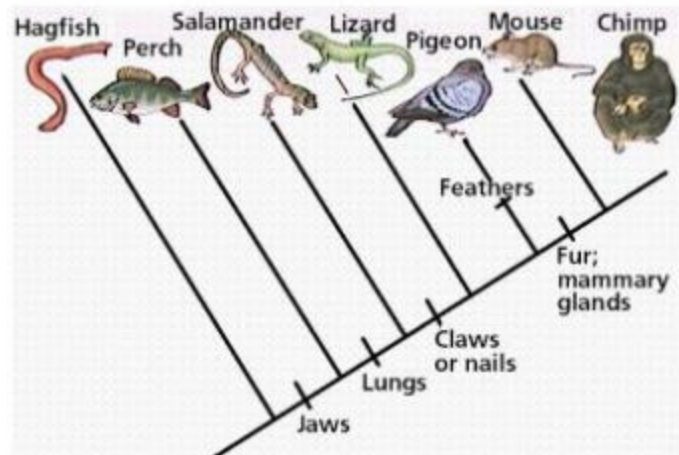
In this phylogeny, of how many clades is species T a member?





1. Which node represents the most recent common ancestor of terminal taxa B and C?
2. Which node represents the most recent common ancestor of terminal taxa A and B?
3. Which terminal taxon is B more closely related to, A or C? Explain how you know.
4. Of the cladograms shown below, which one shows a different evolutionary history from the others? (Think “node rotation” or “swap”)

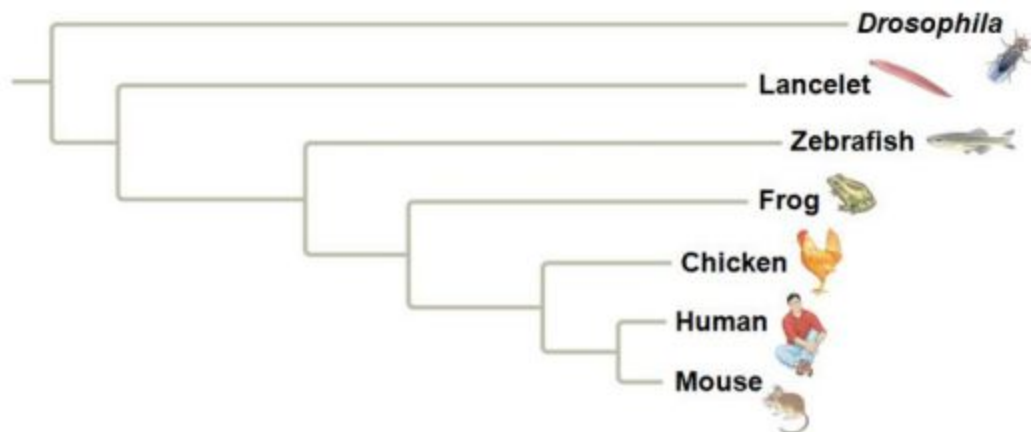


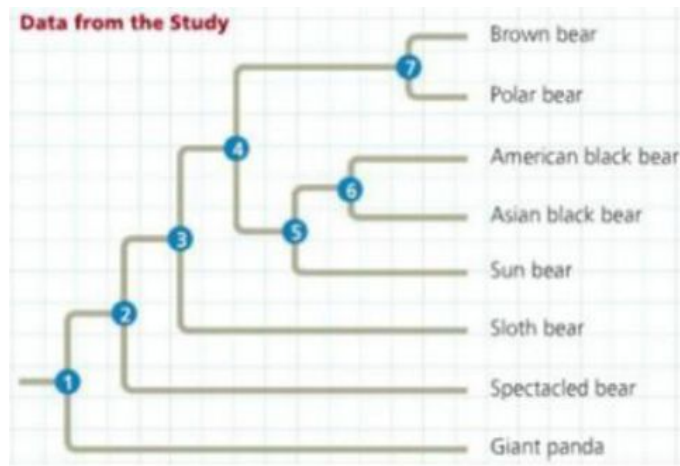


Which of the following groups do NOT form a clade?

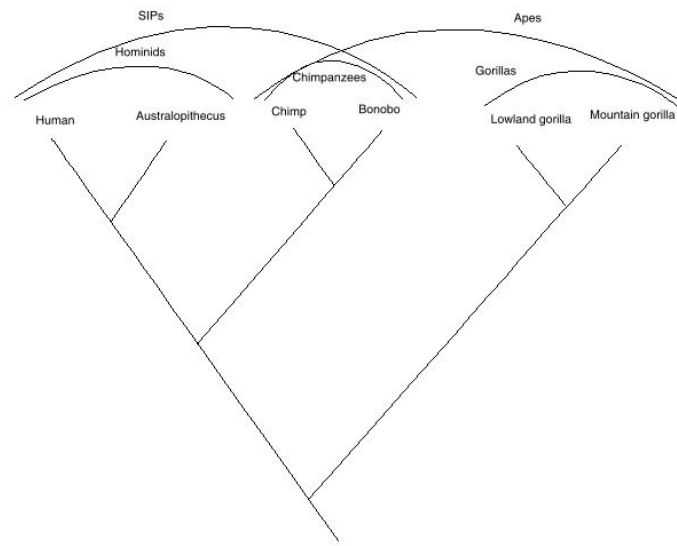
- a. Pigeon, Mouse, and Chimp
- b. Lizard, Pigeon, and Mouse
- c. Salamander, Lizard, Pigeon, Mouse, and Chimp
- d. Hagfish, Perch, Salamander, Lizard, Pigeon, Mouse, and Chimp

14. Using the cladogram below, the relative lengths of the frog and mouse branches in the phylogenetic tree indicate that
- a. Frogs evolved before mice.
 - b. Mice evolved before frogs.
 - c. The genes of frogs and mice have only coincidental homoplasies.
 - d. The homolog has evolved more slowly in mice.





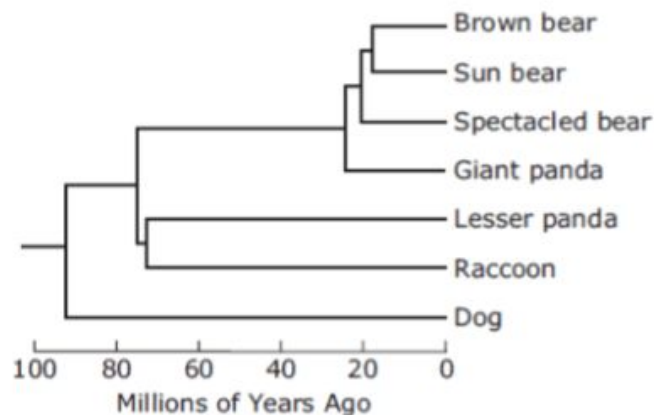
1. Which number represents the most recent common ancestor of (a) all bears, (b) sloth bears and spectacled bears, and (c) the Asian black bear and the brown bear?
2. According to the data represented in this tree, is the sun bear more closely related to the sloth bear or the polar bear? Explain
3. Which species is the outgroup among the bears? What does this mean about its evolution compared with that of the others?



MC. According to the figure above, which of the following groups is *not* a clade?

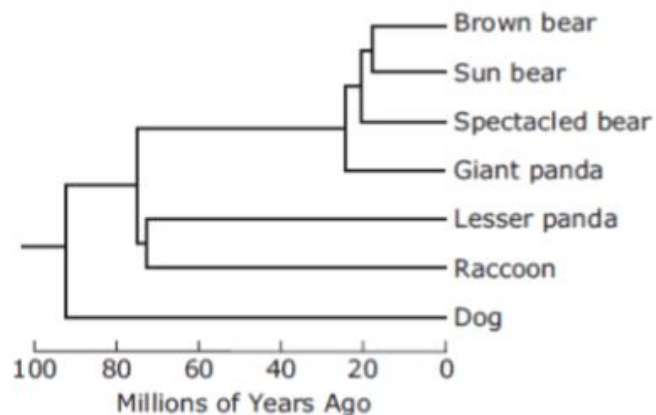
- A. Hominids
- B. Chimpanzees
- C. Apes
- D. Gorillas
- E. Australopithecus

Based on this phylogenetic tree, which statement is true?



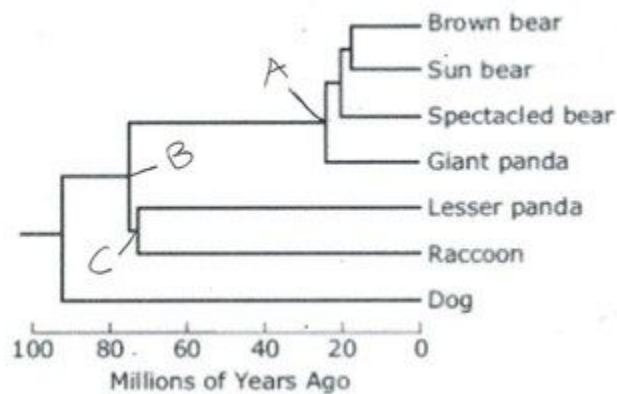
- ☐ Lesser pandas are more genetically similar to giant pandas than they are to raccoons
- ☐ Brown bears are more genetically similar to giant pandas than they are to sun bears
- ☐ Lesser pandas are more genetically similar to raccoons than they are to giant pandas
- ☐ The bear species are all more genetically similar to dogs than they are to giant pandas

Who is the closest relative to the brown bear?



- ☐ Giant panda
- ☐ Dog
- ☐ Lesser panda
- ☐ Sun bear

Which is the most recent common ancestor for the giant panda and the raccoon?



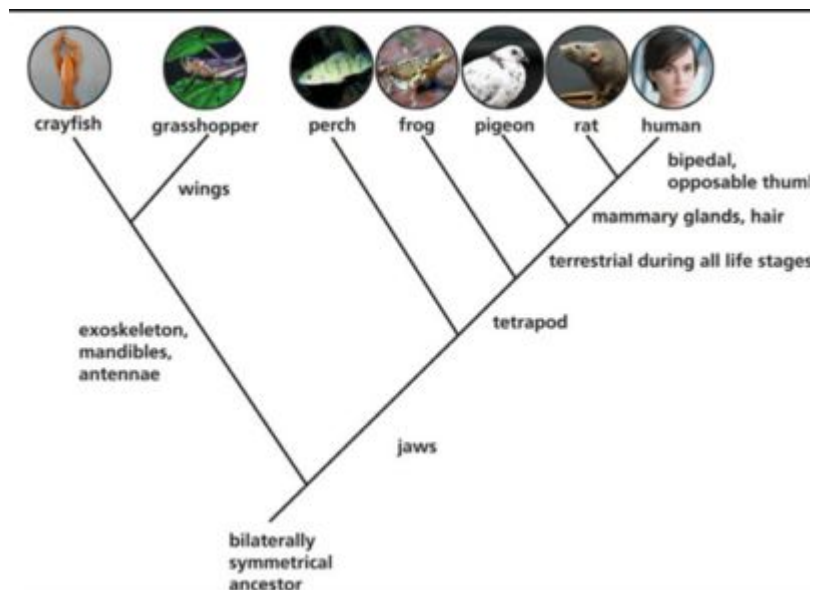
☐ A

☐ B

☐ C

☐ None of these are correct

Which trait do only humans have, according to the cladogram?



- ☐ jaws
- ☐ mammary glands & hair
- ☐ terrestrial during all life stages
- ☐ opposable thumbs